AMP Dock

Sheets

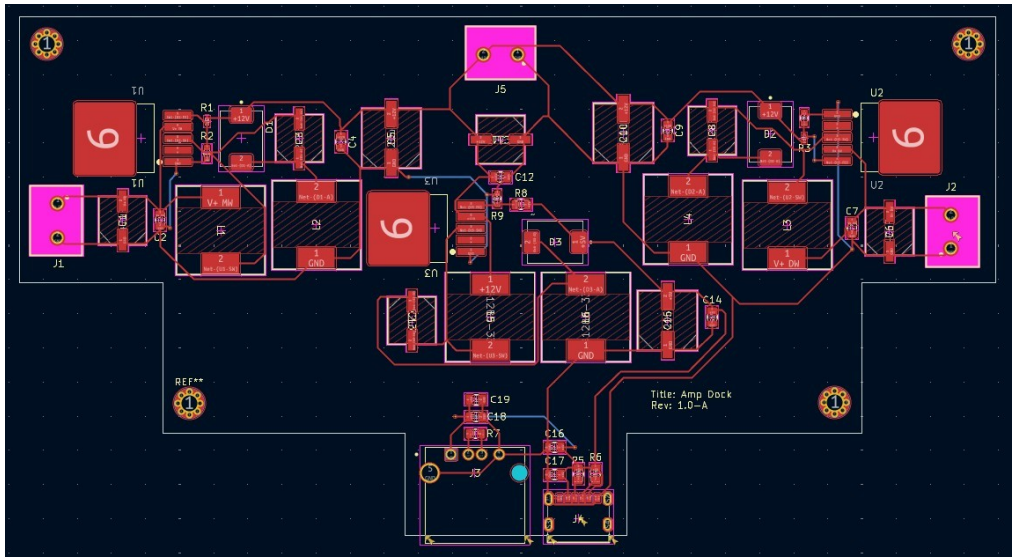
Converter_MW

Outputs

Converter_DW

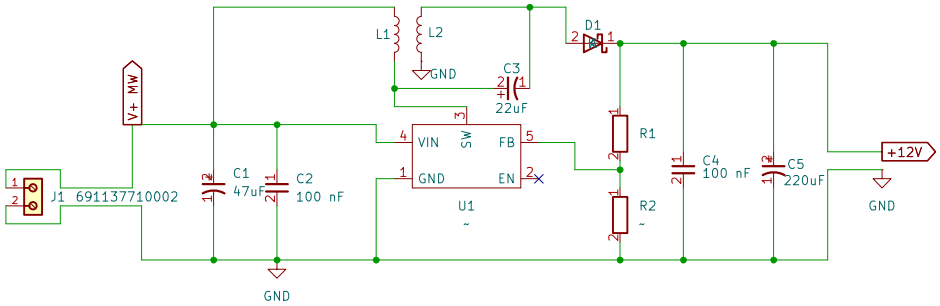
12V–5V Converter

PCB Layout



Description:
The AMP Dock is a power conversion circuit that consists of two different buck converters that take battery inputs from a Milwaukee M18 and a DeWalt XR.
The system steps the voltage of these batteries down to feed a 12 V bus that is used to output 12 V to power a fish finder device or other similar 12 V powered device. The 12 V bus is also parrellelly connected to a seperate buck converter that steps down to 5V for USB-A and USB-C outputs for phone charging. This device will be contained in a IP67 container that has mounting points for the batteries and USB output ports.

Wilson Electrical Design		
Sheet: /		
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Title: AMP Dock		
Size: A4	Date: 2025-10-27	**Rev: 1.0-A**
KiCad E.D.A. 9.0.4		Id: 1/5



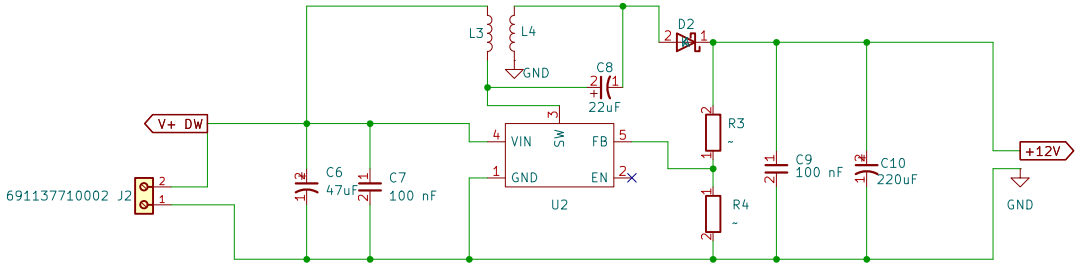
Wilson Electrical Design

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Size: A4	Date: 2025-10-27	**Rev: 1.0-A**
KiCad E.D.A. 9.0.4		Id: 2/5

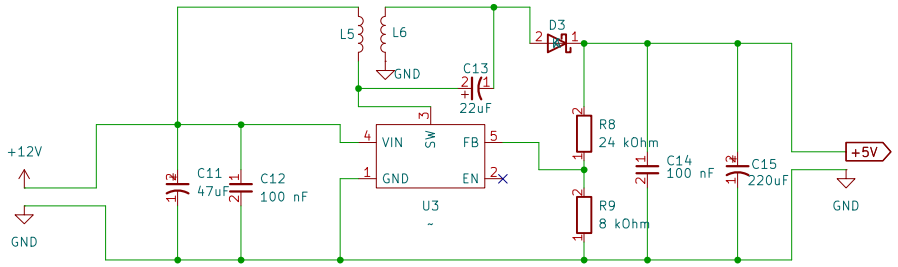


Wilson Electrical Design

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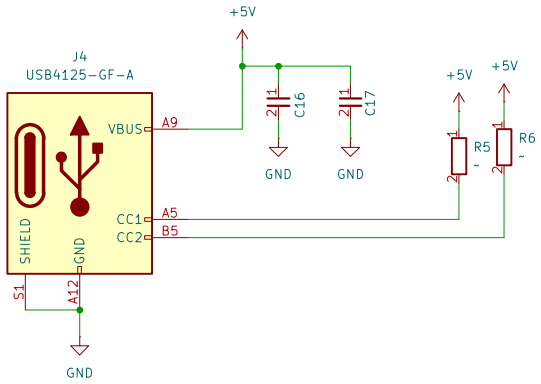
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KiCad E.D.A. 9.0.4		Id: 3/5

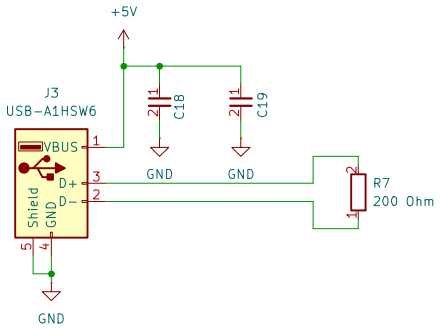


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Title: DC–DC Converter		
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KiCad E.D.A. 9.0.4		Id: 4/5

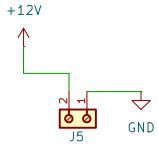
USB-C OUTPUT



USB-A OUTPUT



12V OUTPUT



Wilson Electrical Design		
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Title: Outputs		
Size: A4	Date: 2025-10-28	**Rev: 1.0-A**
KiCad E.D.A. 9.0.4		Id: 5/5